HUMAN-AI CO-CREATION IN FILM MUSIC:
A Practice-Based Study of Creative Agency and Gestural Scoring

Draft Research Design/Plan • ARDN800 Te Ao Mahora / Creative Practice Research Design • 2026

KEYWORDS & METHODS

Human-AI Dialogue · Film Scoring · Gestural Interaction · Relational Agency · Practice-Based Research · Embodied Cognition · Mana · Hā · Whakapapa[1]

DRAFT RESEARCH QUESTIONS

Primary Question: How does gestural interaction shape relational agency in human–AI film scoring practice?

Sub-question 1 — Embodiment & Breath: *How does gestural input influence the perceived sense of embodiment (Hā[2] / Qi[3]) in AI-generated film music?*

Sub-question 2 — Cultural Sovereignty: *How can gestural-AI systems be designed to preserve culturally specific musical structures (e.g., pentatonic, non-metrical forms)?*

Sub-question 3 — Narrative Authority: *How is narrative control negotiated between composer and AI during live gestural scoring?*

Fig. 1 Produced by Ableton Live Studios. Jingrui Han, 2025.

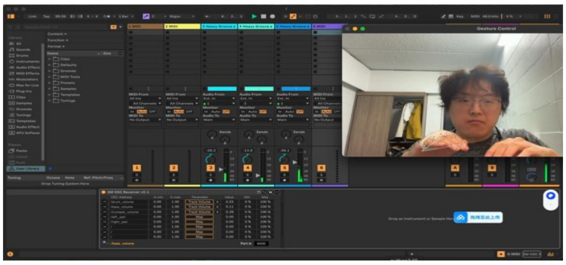

Fig. 2 Ableton Live Gesture Mapping Interactive Design. Jingrui Han, 2025.

AIMS, CURIOSITIES & MOTIVATIONS

As a Chinese composer and PhD researcher at AUT, trained across composition, sound design, and creative technologies, I come to Mātauranga Māori as an outsider who recognizes resonances with Chinese philosophical traditions, and who commits to engaging its concepts responsibly — as a dialogical lens rather than cultural decoration.

My research begins from a persistent unease: Generative AI tools are increasingly automating film music composition, compressing workflows that once took weeks into hours and raising urgent questions about composer displacement[4]. That means composers' Mana, their power, how they help with shaping meaning, is slipping away. A central question is whether such systems can function as genuine dialogical collaborators, or whether they inevitably operate as sophisticated pattern-replicators, producing output without embodied experience or cultural attunement

Most generative AI music systems focus heavily on western tonal styles - almost ignoring pentatonic traditions and non-metrical rhythms I've known since childhood[5]. This representational asymmetry is not merely a technical

[1] Sidney M. Mead and Hirini Moko Mead, *Tikanga Māori: Living by Māori Values* (Huia Publishers, 2003).
[2] Mead and Mead, *Tikanga Māori*.
[3] Dainian Zhang, *Key Concepts in Chinese Philosophy*, with Internet Archive (New Haven, Conn. : Yale University Press, 2002), http://archive.org/details/keyconceptsinchi0000zhan.
[4] Jie Huang et al., "Generative AI in the Screen and Live Performance Industries: A Conceptual Framework and Prospects for Future Research," *Convergence* 31, no. 6 (2025): 1971–2005, https://doi.org/10.1177/13548565251383409.
[5] Atharva Mehta et al., "Music for All: Representational Bias and Cross-Cultural Adaptability of Music Generation Models," arXiv:2502.07328, preprint, arXiv, May 6, 2025, https://doi.org/10.48550/arXiv.2502.07328.

limitation — it constitutes what Hayles identifies as a structural power dynamic: a systemic determination of whose sonic knowledge is treated as computationally legible, and whose is rendered invisible by the training pipeline.[6] Data sovereignty — the composer's right to determine which cultural memories an AI system learns from — is therefore an ethical imperative, not an optional design consideration. This research asks whether embodied gestural interaction can restore what Mātauranga Māori names Hā[7] and Chinese philosophy names Qi[8] — the animating vital presence that distinguishes living creative agency from automated pattern reproduction. Drawing on Ihde's post-phenomenology,[9] the central question becomes: can the composer's body — hands, breath, postural weight — remain an irreducible authorial presence within a co-created score, or does the logic of the generative system always ultimately determine the outcome?

The aim is not to resist AI but to map the conditions under which human-AI collaboration in film scoring can be understood as genuinely dialogical — where, as Bohm argues, meaning emerges between participants rather than being unilaterally generated by pre-trained systems. [10] This research contributes to practice-based research, culturally responsive AI design, and cross-cultural creative technologies scholarship by positioning non-Western musical knowledge as a design constraint and modelling responsible engagement with Mātauranga Māori concepts as an outside researcher.

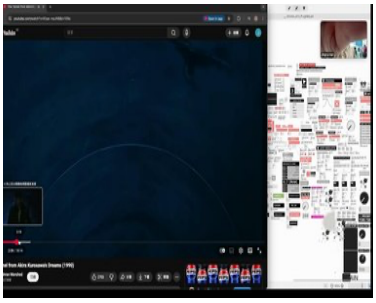

Fig. 3 Kurosawa's Dreams (1990) as emergent narrative model; PD patch. Jingrui Han, 2025.

Fig. 4 Field recording in Beijing. Jingrui Han, 2025.

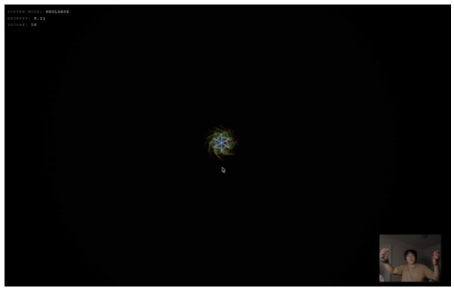

Fig. 5 Gesture Interaction Web. Jingrui Han, 2026.

METHODOLOGY & METHODS

Methodology (philosophical rationale) and **methods** (concrete practices) form a single iterative loop: making → performing → reflecting → redesigning.

METHODOLOGY — The Philosophical Raft

Primary Paradigm: Practice-Based Research	The primary paradigm. Knowledge is generated through and within the creative act itself — not about practice but from practice.
Theoretical Frameworks: Post-human Relationality	Positions the researcher not as a sovereign subject using a tool, but as one node in a relational network including the AI, interface, score, and film image.
TheoreticalFrameworks:Post-phenomenology	Examines how embodied human experience is mediated and transformed through technological instruments.

PBR works better here than ethnography or grounded theory. The research needs insight from the actual creation process. And the final product becomes the core argument. Post-phenomenology helps interpret what prototypes show. It doesn't just record results. Probably, it offers deeper meaning through lived experience. Each prototype reveals something more than surface features.

METHODS — The Four-Phase Practice Cycle

The four methods form an iterative loop: prototyping generates data, performance generates reflection, reflection informs redesign.

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[6] Nancy Katherine Hayles, *How We Became Posthuman: Virtual Bodies in Cybernetics, Literature, and Informatics* (University of Chicago Press, 2010).
[7] Mead and Mead, *Tikanga Māori*.
[8] Zhang, *Key Concepts in Chinese Philosophy*.
[9] Don Ihde, *Technology and the Lifeworld: From Garden to Earth* (Indiana University Press, 1990), https://philarchive.org/rec/IHDTAT-3.
[10] David Bohm, *On Dialogue* (Routledge, 2013), https://doi.org/10.4324/9780203180372.

01 Iterative Prototyping	**What:** Design and build successive versions of a gestural scoring interface using Ableton Live, Max/MSP, OSC protocols, and motion-tracking input. **Why:** *PBR: each design decision is a theoretical claim about agency and cultural value made tangible in code.*
02 Gestural Mapping	**What:** Systematically map physical gestures (hand position, velocity, breath) to AI generative parameters (timbre, density, harmonic mode) via OSC. **Why:** *Post-phenomenology: reveals how the interface reshapes bodily perception and what musical agency becomes possible or impossible.*
03 Live Performance Testing	**What:** Score and perform short film sequences using the prototype interface in real-time. Test both studio conditions and live-audience settings. Record audio, video, and biometric data where possible. **Why:** *Post-human Relationality[11]: tests how agency is distributed across composer, interface, AI, and score in a live encounter.*
04 Autoethnographic Reflection	**What:** After each performance cycle, write structured reflective journals addressing: moments of creative flow and resistance; instances where the AI led or colonized the emotional narrative; cultural dissonances; and adjustments to the next prototype. **Why:** *Autoethnography + Bohm's Dialogue[12]: treats the composer's subjective experience as data, analysing whether genuine dialogue occurred..*

↺ The cycle repeats: Reflection from Phase 04 feeds directly into revised Prototype design in Phase 01.

ACADEMIC, SOCIAL & CULTURAL FIELDS OF INQUIRY

- **Epistemological foundations**: Posthumanism, Mātauranga Māori relational ontology (whakapapa, visible/invisible knowledge), and Chinese philosophical concepts of qi as embodied vitality form the conceptual basis of this research, framing knowledge as relational, processual, and distributed.
- **Methodological framework**:Post-phenomenology (Ihde)[13], embodied cognition, and Bohm's dialogue theory [14]provide the methodological lens through which human–machine interaction is understood as an embodied, co-emergent and interpretive process.
- **Technical and creative domains**: Generative AI for music, gesture-based human–computer interaction, and film sound composition constitute the practical field in which these theories are tested and materialised.
- **Ethical and cultural considerations**: Data sovereignty, decolonising algorithmic systems, and cultural attribution in AI-assisted composition guide the design and evaluation of all system outputs.

REFLECTION: TE AO MĀHORA WĀNANGA

The Te Ao Māhora Wānanga significantly altered the framework of my thesis. Sione Faletau's translation of Kupesi into digital soundscapes illustrated the potential of non-Western cultural encoding to be embedded in computational systems[15], which led me to reconsider my original belief that algorithms are neutral. This revelation shed light on why I chose Whakapapa [16]as a fundamental idea. The sonic materials I work with Chinese pentatonic modes and orally transmitted, non-metrical rhythmic practices function as carriers of cultural lineage and risk marginalization through processes of data-driven abstraction.

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[11] Hayles, *How We Became Posthuman*.
[12] Bohm, *On Dialogue*.
[13] Ihde, *Technology and the Lifeworld*.
[14] Bohm, *On Dialogue*.
[15] Faletau, Sione, "Presentation on Kupesi and Digital Soundscapes. Te Ao Māhora Wānanga, Auckland University of Technology, Tāmaki Makaurau," 2026.
[16] Mead and Mead, *Tikanga Māori*.

The wānanga further deepened my understanding of visible and invisible knowledge within Mātauranga Māori[17]: gestural performance as a form of visible, embodied practice, and AI generative processes as an invisible infrastructural condition. Dialoguing between these domains has become central to my methodological approach. Concretely, this reorientation produced two specific changes to my research design. First, it prompted me to reformulate Sub-question 2 — originally framed as a technical question about style preservation — into an explicitly ethical question about data sovereignty and cultural attribution. Second, it led me to incorporate autoethnographic journalling as a fourth method, specifically to document the moments in each prototype cycle when the AI's generative logic overrides or flattens culturally specific material: these moments of erasure are now treated as primary research data rather than system failures to be corrected.

Bohm's dialogue theory[18] and Ihde's post-phenomenology [19]together substantiate this position: meaning must emerge between participants, and every interface shapes perception in ways determined by its design.

BIBLIOGRAPHY (CHICAGO-TURABIAN)

Bohm, David. *On Dialogue*. Routledge, 2013. https://doi.org/10.4324/9780203180372.

Hayles, Nancy Katherine. *How We Became Posthuman: Virtual Bodies in Cybernetics, Literature, and Informatics*. University of Chicago Press, 2010.

Huang, Jie, Graham Hitchen, and Safak Dogan. "Generative AI in the Screen and Live Performance Industries: A Conceptual Framework and Prospects for Future Research." *Convergence* 31, no. 6 (2025): 1971–2005. https://doi.org/10.1177/13548565251383409.

Ihde, Don. *Technology and the Lifeworld: From Garden to Earth*. Indiana University Press, 1990. https://philarchive.org/rec/IHDTAT-3.

Mead, Sidney M., and Hirini Moko Mead. *Tikanga Māori: Living by Māori Values*. Huia Publishers, 2003.

Mehta, Atharva, Shivam Chauhan, Amirbek Djanibekov, Atharva Kulkarni, Gus Xia, and Monojit Choudhury. "Music for All: Representational Bias and Cross-Cultural Adaptability of Music Generation Models." arXiv:2502.07328. Preprint, arXiv, May 6, 2025. https://doi.org/10.48550/arXiv.2502.07328.

Zhang, Dainian. *Key Concepts in Chinese Philosophy*. With Internet Archive. New Haven, Conn. : Yale University Press, 2002. http://archive.org/details/keyconceptsinchi0000zhan.

AI Tool Disclosure

Claude (Anthropic, claude.ai, accessed April 2026) was used in the preparation of this research design as a research and learning tool for two specific purposes: (1) to check the spelling and orthographic accuracy of Māori terms including Mana, Hā, and Whakapapa; and (2) to assist in identifying and locating academic sources. All writing, arguments, research direction, and intellectual content are the author's own. Use of AI tools was discussed with course lecturers in accordance with ARDN800 guidelines.

Example prompt used: "What is the correct spelling of Māori terms such as Mana, Hā, and Whakapapa?"

[17] Mead and Mead, *Tikanga Māori*.
[18] Bohm, *On Dialogue*.
[19] Ihde, *Technology and the Lifeworld*.